Cost Management Optimisation

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ABSTRACT

Processing expansion options and mining fleet purchases are major cost items that receive due focus in most cost management analyses. If these items are not needed, they can significantly reduce a project's short-term capital spend and improve profitability and value to shareholders. When conditions change prior to purchasing equipment, the purchase may no longer represent the best decision for the project.

This paper outlines a structured approach for making major cost decisions to improve the value of a project, determining which items should be kept and which should be eliminated.

INTRODUCTION

Managing costs is critical to the operation of any major company, particularly operating mining businesses. While revenues can be difficult to control due to the underlying geology of the deposit or the market determined price, costs are normally within the control of a business. Costs can be kept at previous levels, scaled back, eliminated or increased with high levels of control.

The flexibility to reduce costs has often been used strategically to improve the profitability of the business and, importantly, its value to shareholders. Regrettably though, there are many examples of revenue reduction as a consequence of across-the-board cost reduction. The net impact is substantially reduced project profitability and a destruction of shareholder value. What is needed is a framework to determine which costs are excessive and should be reduced and which costs are not and should be maintained or even increased.

COST MANAGEMENT OPTIMISATION OVERVIEW

A framework for managing costs needs to be designed to maximise value to shareholders. To be effective, this framework needs to incorporate both clear technical and relevant people issues into a practical, repeatable and generic system.

Single overarching objective

The first component of the framework is to clarify a single overarching objective that can be used for a project, incorporating all key elements. While this seems to be a simple step, projects often use multiple objectives or 'proxies' for shareholder value. These proxies need to be checked to see if they do in fact add shareholder value for each scenario being evaluated.

On first inspection, it may seem that total cost could be used to compare different scenarios where lower costs are better. While cost reduction programs have the ultimate goal of improving shareholder value, they might be undertaken using the simple goal of minimising costs. There are often costs that can be identified that are no longer needed, do not impact downstream processes and can be eliminated to simply improve profit and thus the value of a project. Cost minimisation implemented without consideration of downstream implications on the business can result in losing more value than the costs saved, thereby destroying shareholder value.

Before we can evaluate if a cost should be reduced (or increased), we need overarching criteria to measure its value to the shareholders. Before defining the best measure of shareholder value, let's take a moment to consider why there needs to be only one overarching objective, rather than many.

Companies often have multiple goals displayed prominently for employees to read and give their best efforts to achieve or exceed. These may include reducing injuries and unsafe practices, achieving an annual production target, increasing the mineral reserves, being an employer of choice, keeping costs less than industry targets and so on. While these are often well aligned with each other, they can also conflict with each other. For example, when determining the scope and scale of an exploration drilling program, keeping costs down can conflict with increasing mineral reserves. Which measure should be used?



FIG 1 – Many proxies for shareholder value.

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TABLE 1
Multiple conflicting corporate objectives.

Increase metal production to X
Reduce operating costs below the X quartile
Increase the ore reserves to X by year Y
Be the employer of choice for the mining industry
Environmental stewardship and sustainability
Your safety is our priority – zero fatalities and reduce lost time injuries to X

With many objectives, employees may need to undertake analysis for each objective, resulting in similar work being undertaken many times with the associated increased time for making the decision. Once all the work is completed for each objective, there is the problem of which one to use. Ultimately, only one choice can be made for each decision, and normally each objective produces a different result. How do you ensure that the best option for the company is chosen? By choosing the overarching measure of shareholder value, which appropriately balances revenues (eg from all the metals or products), costs (eg exploration, mining, processing, marketing, sales), productivities (eg truck and shovel utilisation and efficiency, metal recoveries) and all project constraints (including mining fleet, processing, infrastructure, safety and environmental).

Net present value (NPV) is widely used in the mining industry (and all other industries) and is normally the best proxy for shareholder value (Brealey, Myers and Allem, 2010). While there are some limitations, it is a far superior overarching measure of shareholder value than individual department objectives and all other financial measures (including payback period and internal rate of return). NPV allows a fair weighting between different departmental costs and capital injections and also recognises the monetary value of time over the life of the project. This inherent balancing of costs, revenues and current and future values has also led to NPV being used by stock market analysts to estimate the share price position of listed companies from all industries, not just mining.

While the impact on NPV is sometimes difficult to measure, it remains the standard instrument that major capital decisions are best measured with. This paper assumes that *NPV is the best measure of shareholder value*. The projects being considered are large, typically over ten years, and need to determine the best capital cost management strategy over the life of the project.

The NPV needs to be maximised over the *entire life* of the project using *realistic periods*. While the casual user often assumes both of these criteria, some algorithms have a '*distorted NPV*' calculation that biases the final choice of the best decision.

For example, when scheduling a 30-year project, if the algorithm only considers five annual periods at a time during the optimisation, then revenue and costs outside of this period are not considered and there would be reduced incentive to strip a phase that does not have its ore mined in the five years, thus biasing related decisions. Compensating by creating artificial minimum stripping 'constraints' creates a partly manual schedule and no longer guarantees the best NPV for the project.

A similar problem occurs when algorithms combine annual periods into 'super periods' for the optimisation. This results in a very practical issue since the ore mined at the end of the period is considered available at any stage of this 'super period', unrealistically overstating the value of such a scenario. While these assumptions may be useful in restricted situations, better decisions are made when algorithms maximise NPV over the whole project life with realistic periods (Lane, 1964, 2014).

Prioritised alternatives

Prioritised activity should not be a surprise for those familiar with maximising the value of a system. This paper prioritises capital costs over operating costs and other productivity decisions. While analysis should be prioritised according to the overarching objective, when starting this process the objective may not have been established and a rule of thumb or expert judgement may be initially used. A simple rule of thumb for determining which capital costs to start with might be to analyse the largest capital expenditure items first.

While operating cost reduction programs typically reduce costs by tens to hundreds of thousands of dollars, capital cost decisions regularly involve many millions or even billions of dollars. Although both types of decisions can be worth implementing, shareholders want to prioritise the decisions that are likely to make the greatest impact on the value of a project.

For capital costs, a list of possible capital reduction and expansion options will normally be evaluated. This often includes both the size and timing of fleet expansion and processing capacity options, each with associated capital, operating cost and productivity implications. Brainstorming with a group of expert and senior leadership staff is often useful for identifying a range of alternatives and their possible impact on project value. This can be used for the analysis of first-order prioritisation.

This framework will become an iterative one where measurements are made that can be used to prioritise the alternatives. As the process is followed, it naturally promotes the priority of high-value choices over low-value ones.

Prioritising tasks are only beneficial if action is taken that follow these priorities. If managing large capital decisions is identified as the most important priority in the current environment, it must be followed by appropriate analysis and action on this priority.

Measurement and quality optimisation

Now that we are acting on our prioritised alternatives, we will need to measure the impact of the various options using our objective. As capital expenditure often has many implications on the mining and processing schedule, a new schedule is normally needed to determine the new NPV. This is often not obvious to financial analysts who are not familiar with the mining process, as illustrated in the following example.

Consider a scenario where the cost of various mining fleet capital is higher than previously expected (lower prices can produce the same effect). The tendency is to simply use the same schedule and fleet requirements and change the capital cost when the new equipment is purchased. If the costs are significantly higher, some of the mining stages will not be profitable and the analysts may simply 'lop off' periods at the end of the schedule that are not profitable. The resulting schedule may still include some unnecessary stripping and will probably use cut-off grades that are no longer appropriate for the shorter life project. A better solution can often be found by optimising the schedule with the new costs being considered. In a revised and optimised schedule that considers these new capital costs, the stripping for phases that are not mined is completely removed and cut-off grades for each period are recalculated to maximise project NPV. The

result is often higher-value schedules with a more realistic valuation of the scenario, leading to better decisions.

The quality of both the NPV calculation and the optimisation algorithms are critical to measuring project value and the resulting decisions that are made using this information.

Analysis and feedback

Now that we have high-quality measurements of the value of the various capital investment alternatives, we need to undertake some analysis to determine the best path forward.

Much of this analysis is very straightforward if you can simply look at the range of NPV measurements for the various options and pick the scenario with the highest value. However, some caution should be exercised at this point to ensure that those undertaking the analysis are appropriately trained. For example, we may have considered a range of expansion options for the mining fleets (trucks, shovels and associated ancillary equipment). We may have also considered a range of processing capacity expansions with their associated impacts on costs, recovery and productivity. An experienced strategic planning engineer would ensure that some cases of increased mining capacity are undertaken with increased processing capacity.

One of the great advantages of the analysis stage is learning about where the significant value drivers are for a project. Often, there are very few people who have access to information on the value of an entire project so there are *many new areas for improving the project value discovered during this stage*. The importance of learning the value drivers can be further leveraged by ensuring that site staff undertake the analysis. Site staff that 'live and breathe' a project are often better placed to respond to changing conditions and possess greater knowledge of what can be changed to help improve the project value.

Analysis and feedback is best undertaken by a multidisciplinary group of people. For example, a change in the production schedule would be provided to mining operations, processing, logistics, marketing, finance and management for feedback. While all cases cannot be provided to all people, the more important cases should receive greater review to pick up mistakes, gain greater team ownership and uncover more ideas on ways to further improve project value.

Some periodic independent and expert feedback into the analysis is considered best practice given the magnitude of the decisions and the prevailing levels of experience in this field in the industry. The independent feedback would ideally have experience in the optimisation techniques discussed in this paper across a broad range of projects.

The analysis will typically result in some additional scenarios to evaluate or the fine-tuning of decisions. This feedback should flow back into the new prioritised alternatives, better measurements and sometimes even into the objectives to make sure that it truly reflects shareholder value.

CASE STUDIES

Three cases have been provided to help see how this cost management optimisation can be implemented. These case studies are from areas where large capital expenditure is often incurred: mining fleets, extensions to mining areas and process plant expansion. The final example is then extended to discuss how substantial additional shareholder value can be realised without any significant capital.

Mining fleet capital

The largest capital items in the mining side of the operation are normally the truck and shovel fleets. Additional trucks and shovels are often required to expand the fleet capability or replace old equipment. The planned size of the mining equipment fleets is often determined in the feasibility study for the project based on the geology, mining, processing, environmental and market costs, productivities and prices. As many of these factors can change over time, the appropriate size of fleets needs to be reviewed periodically, typically annually.

For example, consider the size of the truck fleet. The number of trucks is normally matched to the number of shovels; the cycle times to the various waste dumps and crushers.

Firstly, consider the implications on the truck fleet of a simplistic goal of reducing annual costs by ten per cent. The cost reduction will normally also carry an associated goal of maintaining the budgeted metal production so as not to lose revenue. To reduce the annual cost of the operation, projects can reduce the number of trucks operated, perhaps park ten per cent of the trucks and/or not buy additional trucks. In order to not sacrifice the metal production while using fewer trucks, plans can be adjusted so that the areas containing ore continue to receive high priority, but the waste handling receives a lower priority. In this way, metal production targets are still achieved and the operating costs of the business are reduced. The truck fleet requirements could be substantially reduced in this scenario. The waste that is no longer mined may not impact on the business for several years until eventually the reduced availability of high-value ore is impacted. In this scenario, the impact on shareholder value can be substantially worse than the costs saved, while the high-capital processing assets and business infrastructure are not utilised with high-revenue-generating ore. We clearly need a much more sophisticated measure of shareholder value than the cash flow for a few years.

The implications on the value of the project for large capital decisions last many years or even decades. This framework will use the full value implications on the entire business (NPV, the single overarching objective), not just during the budgeting period. Analysis of the project for scenarios with and without the extra trucks (finding a profitable case for a good decision) requires multiple cases to determine the size and timing of the truck fleet.

The impact on revenue and costs also needs to be evaluated according to the year in which the value is realised. Although costs saved this year are more valuable than the same revenue lost next year, substantial revenue forfeited in the future can outweigh short-term cost savings.

This systematic analysis leads to many measures of potential project value that will then reveal if cost reduction goals should be applied to the truck fleet and how large the cost reduction should be. If the delayed or reduced revenue is less than the value saved through cost reduction, should this decision be implemented? There may be times when significantly more than ten per cent of costs should be saved, and other times when costs should actually be increased so that maximum project value is achieved.

Prestripping is a clear example of when we need to incur upfront costs to gain revenue in future years. Large multiphase projects require many stripping initiatives, and each needs to be implemented to maximise shareholder value. These large projects require the cash flows from every period to be optimised simultaneously so as to maximise the value of the project and associated shareholder value.

Mine extension or closure

As a mine matures, it is common for additional reserves to be included in the project or for parts of it to be closed. Extensions may be in the form of satellite projects, additional pushbacks and/or the inclusion of underground operations as the surface mining draws to an end. The capital costs for mine extensions may include additional workshops, mining fleets, conveyor extensions and services. Closure of mines or parts of mines has the potential to reduce costs significantly.

The single overarching objective remains to maximise the value of the entire business. When capital is limited and the economics of projects are dropping, the temptation can be to cut those projects with the lowest rates of return or the longest payback periods. Both of these financial instruments have serious flaws (Brealey, Myers and Allem, 2010) and we would be better to determine the NPV for the different options. A set of cases with a range of capital requirements should be optimised to measure value (NPV) from each scenario.

Care is needed to ensure that each case is compared with similar optimisation rigour and that it can be implemented in practice. We should not put a lot of energy into tweaking the value from our favourite case since this can lead to bias in the results. All cases should be optimised to a similar level of rigour so that they can be compared fairly and decisions made that truly improve shareholder value.

During downward cycles, companies tend to quickly spread the message that there is 'no capital' available during these times. However, these statements rarely actually mean 'zero' capital constraint. Instead, they typically mean that significant justification will be needed for capital requests, and the larger the request the greater the justification. If there truly is a capital limit, then we should undertake the bulk of our analysis within this constraint and be extremely careful of unrealistically 'unconstrained' scenarios.

Process plant expansion capital

Processing plant expansions can be a source of substantial value to a long life project. An 80-year project has many years that have a very low impact on the project value. The impact of ten per cent annual cash flow discounting means that revenue delayed by two years only receives 81 per cent of its earlier value. After ten years, this drops to 35 per cent, and after 44 years it is less than one per cent. This provides a significant incentive to delay large costs and a strong motivation to bring forward revenue. Many analysts therefore only consider the first 20–30 annual cash flows to value mining companies. This discounting of cash flows encourages processing plant expansions to be considered since they may move revenues from years 40–80, where they contribute little to project value, into the first 20–30 years.

We start with the objective of maximising shareholder value and NPV as the best proxy for this. This is an important first step since it is easy to become distracted with the additional metal that is produced and lose sight of the costs required to produce it.

We now have a range of processing expansion options to consider. Doubling and tripling the processing capability has advantages in terms of simplicity of design, implementation and maintenance, but does not necessarily provide the highest value for shareholders. There may also be good reasons to use different equipment or only expand part of the processing capacity.

When evaluating each scenario, we want to make sure that we make the most of each capital step. For example, the mining policies such as pushback sequencing, stockpiling and cut-off grades should be optimised simultaneously for each scenario. Processing policies such as throughput/recovery relationships should also be simultaneously optimised with the mining policies for each case. As a further example, a copper flotation circuit may have its throughput and recovery changed by optimising the semi-autogenous grind size and flotation residence time simultaneously with the mining policies. Projects using leaching may vary leach times, irrigation and oxidation rates together with the mining policies.

Simultaneous optimisation of mining and processing policies is an old concept that has wide application for determining which processing plant expansion should be selected. If mining and processing policies are not optimised simultaneously, then lower value schedules result and decisions can be made that significantly destroy shareholder value.

These policies are set to achieve maximum recovery 'proxy' to the maximum shareholder value. This proxy can hide significant value to the project when optimised using full project NPV. We can also optimise this policy over time; we normally see higher throughput rates early in the mine life when ore is plentiful, and maximum recovery in the last few years when the mine is shut down if it does not extract every economic part of the revenue-generating products. A key optimisation benefit is often found when the *processing policies are varied over time with the mining policies*.

PROCESS PLANT EXPANSION BENEFITS WITHOUT CAPITAL

The rigour of systematic optimisation should also be applied to the base case where no significant capital is required. For example, the integration of throughput/recovery relationships with mining policies (cut-off grades, phase sequencing and stockpiles) normally provides a substantial improvement (five to 15 per cent NPV) in the value of a project. Rather than paying large amounts of capital for new processing facilities, similar improvements in metal production and revenue can be gained by using higher throughput rates with slightly lower recoveries. Some additional mining equipment may be required to realise this value, which may be very cheap compared to the expanded processing plant option.

The capability to undertake this analysis is now available commercially, although much of the industry is unaware how substantial the value is, how low the capital costs are to implement it and how easily the analysis is to undertake. These techniques have been applied to coal, copper, diamond, gold and poly-metallic mines, and it is likely that there is application for other commodities through investigation.

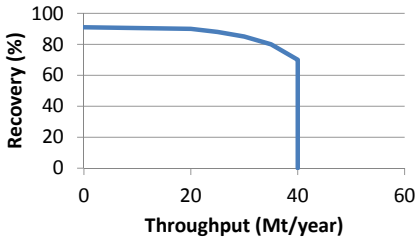


FIG 2 – Throughput recovery processing relationship.

CONCLUSION

Costs are a vital component of project value and, unlike the revenue and discount rate, we have substantial control over them. To reduce costs without destroying shareholder value, we need to model the impact of these cost reductions against changes to the value of the project. The crucial starting place of the cost management optimisation is the identification of the overarching objective that aligns the analysis with shareholder value – normally project NPV. By defining the criteria to be measured for each case, we greatly increase the efficiency of analysis and the value delivered to shareholders.

Optimisation also has a crucial role in making sure that decisions are in the best interest of the company. Care must be exercised to make sure that full NPV calculations are used in the optimisation and that simultaneous policy optimisation is undertaken for decisions.

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